Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

"To eliminate the Dynol Gun & lift failures CAI yard area"

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio –Xylo TCF.

	Plan	Actual
Start Date	**05.06.19**	**22.06.19**
Completion Date	**28.10.19**	**28.10.19**

Current Status: **CLOSED**

Cross Functional Team

No	Description	Name	Designation	Dept	Phone
1	**Team Leader**	Sagar Kekade	L-8 O	STCF	8237028924
2	**Mentor**	Tushar Patil	L-5 M	STCF	9922409136
3	**Member 1**	Sudhir Dhanokar	L-5 M	STCF	9049992381
4	**Member 2**	Pankaj Tandle	L-7 O	STCF	9922944452
5	**Member 3**	Dattatray Jondhale	L-7 O	STCF	7709341353
6	**Member 4**	Bhamre Sachine	L-7 O	STCF	9158960111
7	**Member 5**	Satish Patil	L-7 O	STCF	9922448571

Step1. Select Subject

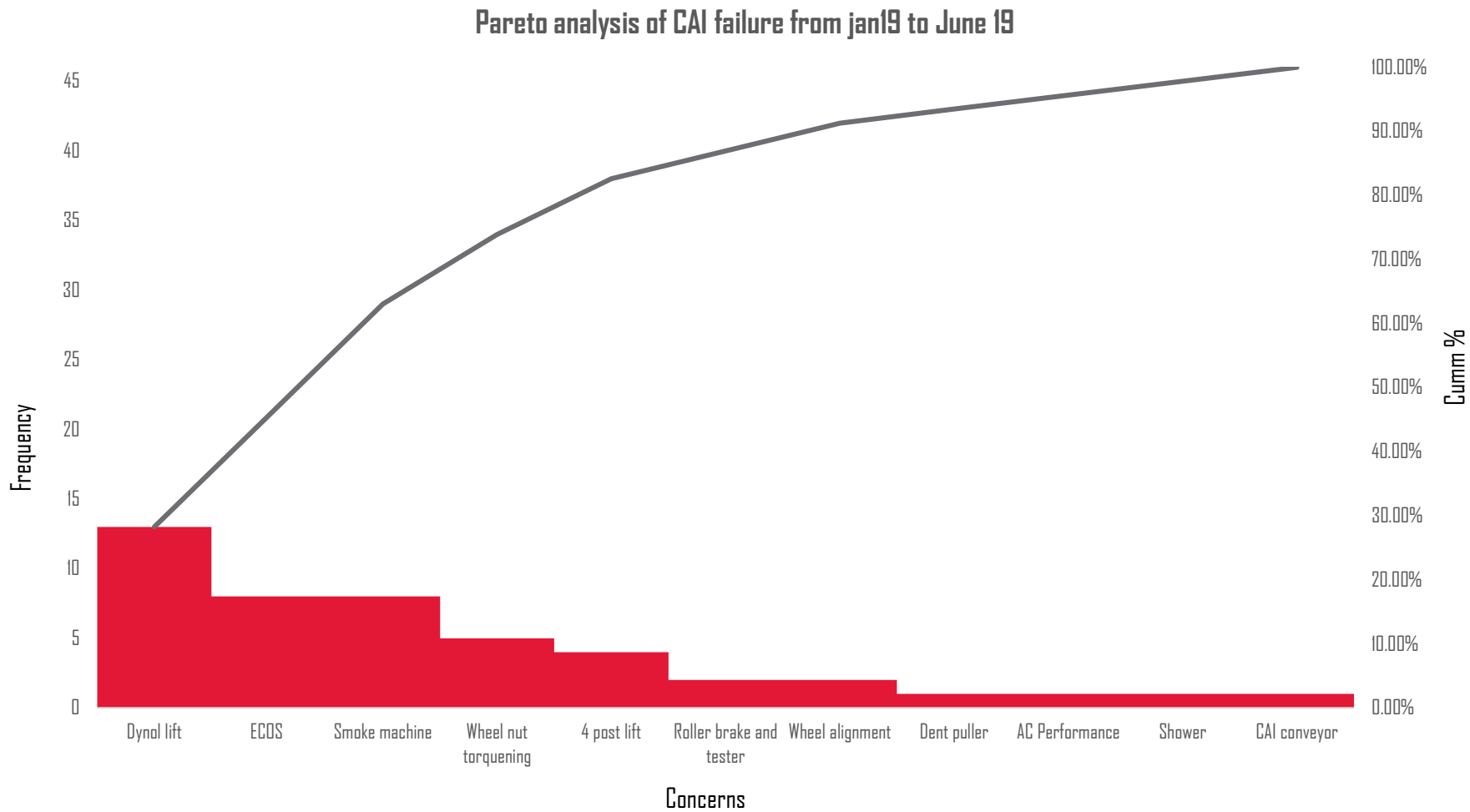
PROJECT TITLE :-
To eliminate the Dynol Gun & lift failures
CAI yard area

Brief Description :

- **What** : Dynol application not working properly.
- **Where** : At CAI yard, Dynol Application
- **When** : Jan-19
- **Who** : By production associate .
- **Why** : scissor lift continuously problem while working
- **How** : Avg 15 inidences in month

Mechanical stopper Not Working properly

2. Justify the choice



<u>INFERENCE :-</u> AS PER PARETO GRAPH DYNOL LIFT PROBLEM IS TOP CONCERN IN MINOR STOPAGGE.

2. Justify the choice (reasons)

PROMISE 2022
AUTOMOTIVE SECTOR

We Will Grow

Revenue 75% | Profit 100%

By

'Enhancing lifestyle' in personal segment
'Enriching livelihood' in commercial segment

Being #1 in UV, SCV, EV and #3 in HCV
Doubling volume from overseas markets

Leading in next generation mobility solutions

#InspiringPeopleToRise

Crusade 2.0

Departmental Objective :-
BIQ – DPR Improvement
- FRC / FBO Improvement

Step 3. Understanding Current Situation

Vehicle Dynol Application

Vehicle Enter in Dynol booth

Vehicle lifted up by Dynol scissor lift

Dynol application through gun

Vehicle lifted down by associate

DEFECT DETECTION STAGE

Vehicle Check

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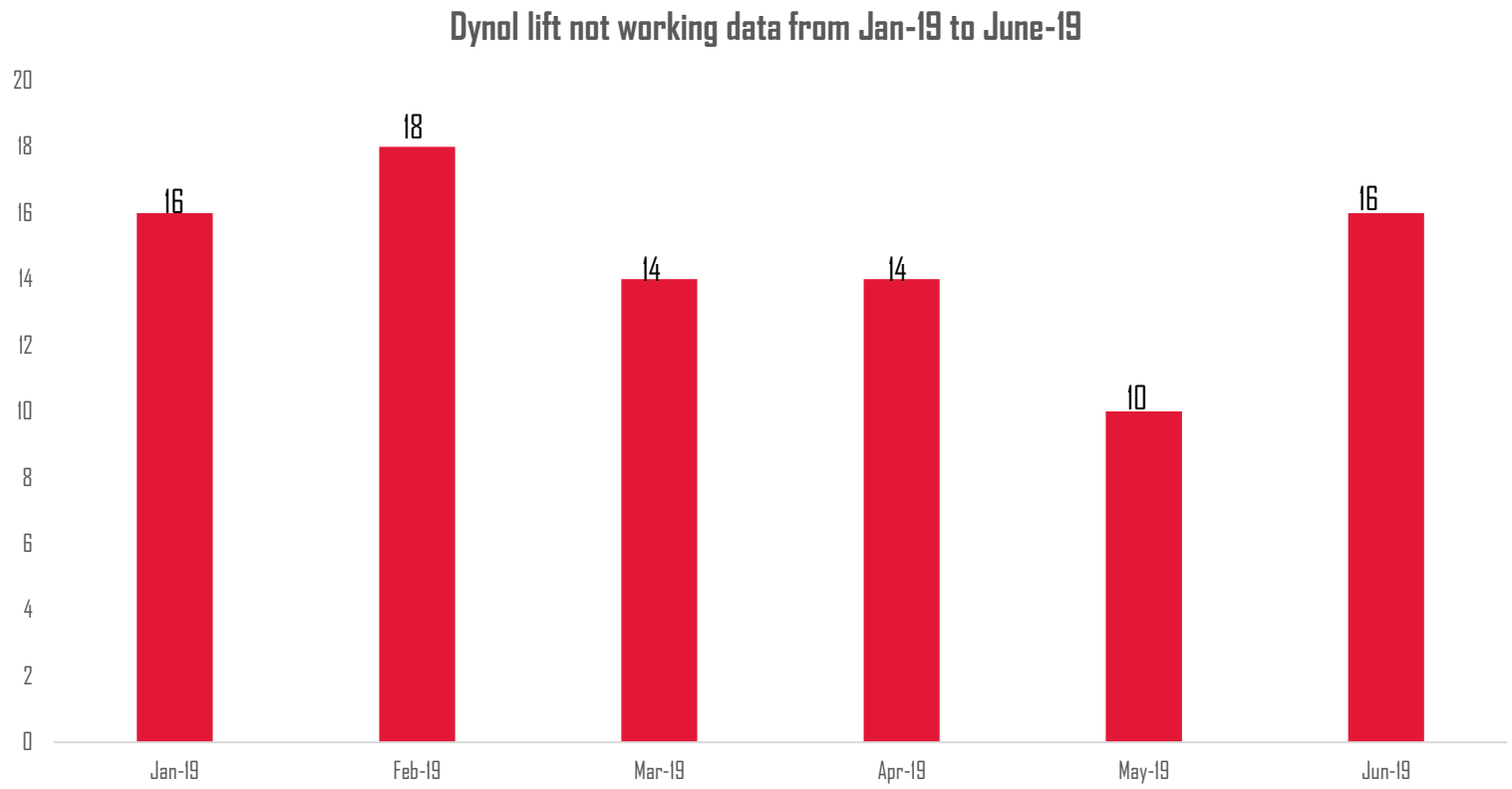
3. Understand the current situation

Month wise Trend



<u>INFERENCE :-</u> AS PER MONTH ON MONTH DATA COLLECTED THE SIDE DEFECT IS CONTRIBUTING AVERAGE 15 concerns/month which causes vehicle loss per month

Step 3. Understanding Current Situation

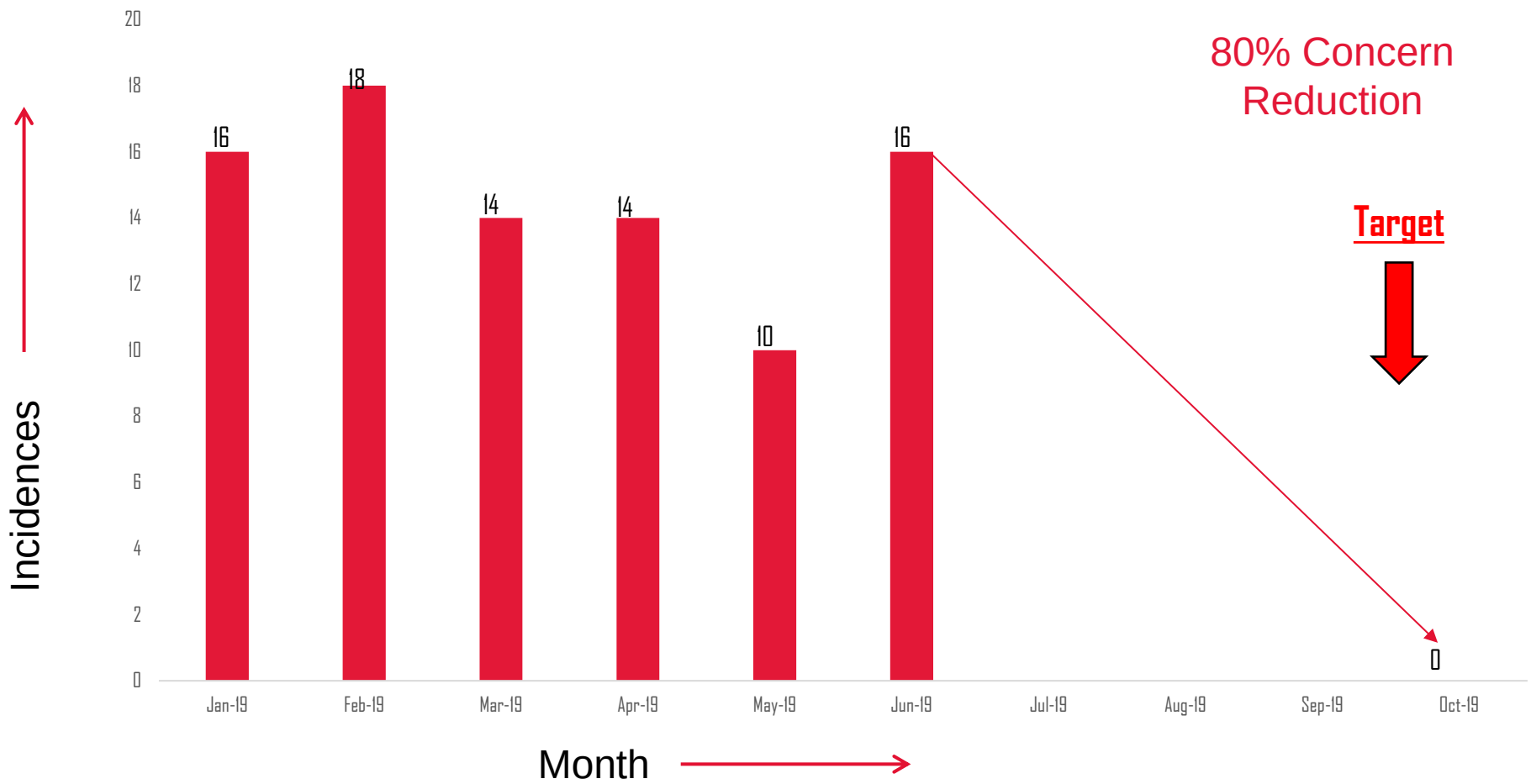
Good observations

- Analyzed after vehicle entered in Dynol booth.
- Scissor lift not down if limit switch not operate which is provided on mechanical stopper.
- Limit switch provided on both cylinder is not operate if mechanical stopper not opened and hence lift not goes down.
- Issue Resolve within 10 to 15 min.
- Issue reported in both the shifts
- Issue occurs frequently
- Reasons for stoppage is similar

4. Select targets

Present level :- 54 cases reported in last 06 months

Target :- Concern elimination by END OF OCT-2019



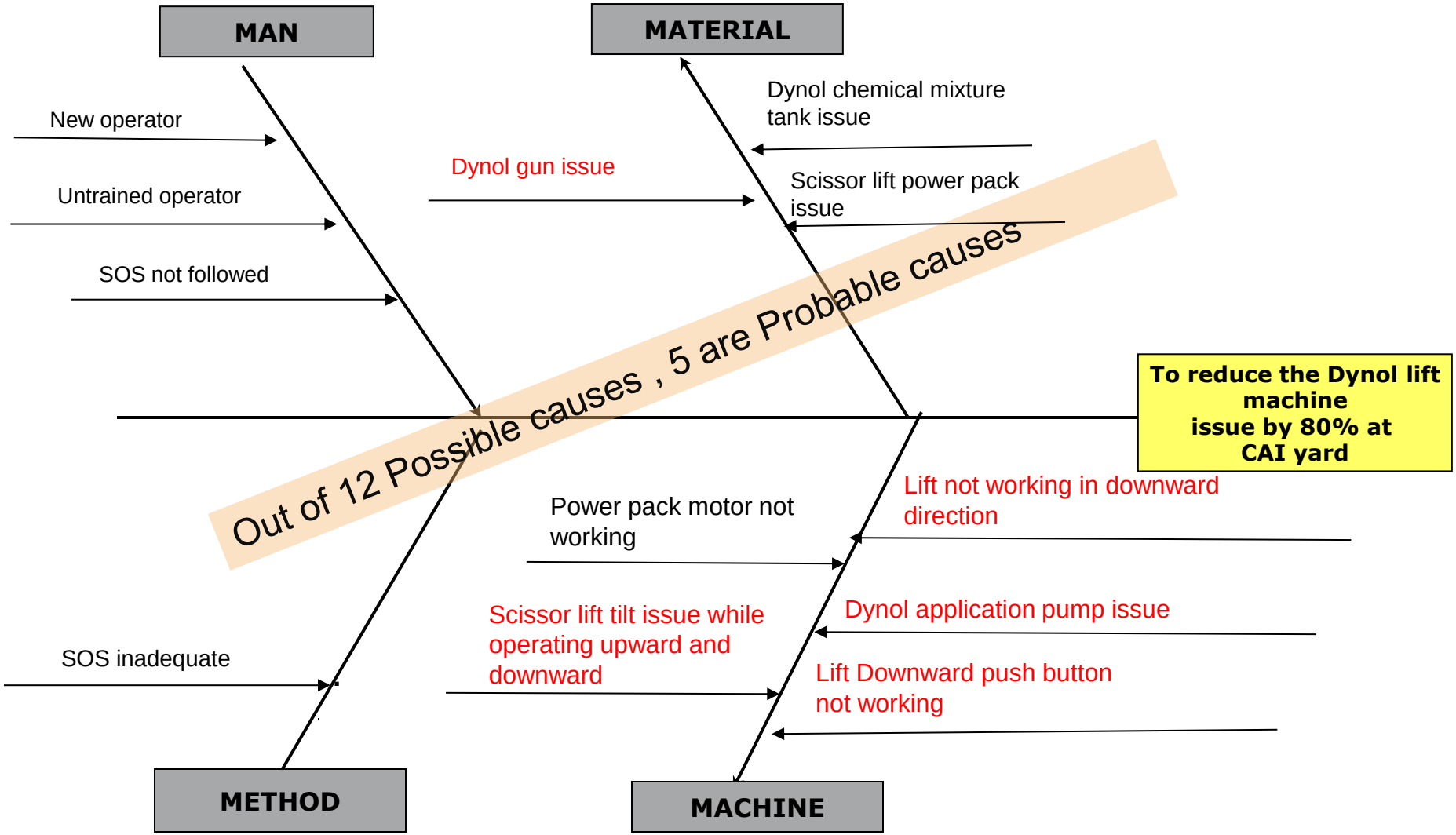
Step 5. Analysis

List of probable causes identification in Brainstorming session

Sr. no	Possible cause	Sr. no	Possible cause
1	New operator	7	SOS inadequate
2	SOS not followed	8	Untrained operator
3	Scissor lift tilt issue while operating upward and downward	9	Dynol gun issue
4	Dynol chemical mixture tank issue	10	Dynol application pump issue
5	Dynol lift power pack motor issue	11	Lift Downward push button not working
6	Scissor lift power pack issue	12	Lift not working in downward direction

Step 5. Analysis

List Of Possible Causes



Step 5. Analysis

List of Probable causes identification in Brain storming session

Sr no	Possible Causes	Dattatray	Tushar	Sudhir	Pankaj	Dattatray	Sachin	Satish	Total
1	New Operator	1	0	0	0	3	3	3	10
2	Untrained operator	3	3	0	0	3	3	3	15
3	SOS not followed	3	3	0	0	3	1	0	10
4	SOS inadequate	9	0	0	1	3	0	0	13
5	Dynol chemical mixture tank issue	3	3	0	0	3	3	3	15
6	Dynol lift power pack motor issue	3	0	0	0	3	0	3	9
7	Dynol gun issue	9	9	3	9	3	9	3	45
8	Scissor lift power pack issue	3	3	3	3	1	0	0	13
9	Scissor lift tilt issue while operating upward and downward	3	9	3	9	9	3	9	45
10	Lift Downward push button not working	9	3	9	9	3	3	3	39
11	Lift not working in downward direction	9	3	3	9	3	9	9	45
12	Dynol application pump issue	9	3	9	9	3	3	3	39

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting cause for hypothesis is 39.

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering

Sr. no	Not considered Causes	Reasons
1	New Operator	NO New operator deployed on this stage
2	Untrained operator	All trained operator
3	SOS not followed	SOS followed
4	SOS inadequate	SOS was OK
5	Dynol chemical mixture tank issue	Mixture tank fan and heater working OK
6	Dynol lift power pack motor issue	Motor working OK
7	Scissor lift power pack issue	Power pack OK no oil leakage

5.1 Identify Probable cause

No	List of Probable Causes
1	Dynol gun issue
2	Scissor lift tilt issue while operating upward and downward
3	Lift Downward push button not working
4	Lift not working in downward direction
5	Dynol application pump issue

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Dynol gun issue	Check gun spray and no gun leakage	Observed gun spray is more and due to this dynol chemical leakage from gun.	**Hypothesis valid**

Equipment Name	Possible Causes	19-Apr	19-May	19-June	19-July	19-Aug	19-Sep	19-Oct	19-Nov	19-Dec	19-Jan	19-Feb	19-Mar
	New Operator	0	0	0	0	0	0	0	0	0	0	0	0
Dynol application	Untrained operator	0	0	0	0	0	0	0	0	0	0	0	0
	SOS not followed	0	0	0	0	0	0	0	0	0	0	0	0
	SOS inadequate	0	0	0	0	0	0	0	0	0	0	0	0
	Dynol chemical mixture tank issue	0	0	0	0	0	0	0	0	0	0	0	0
	Dynol lift power pack motor issue	0	0	0	0	0	0	0	0	0	0	0	0
	Dynol gun issue	2	3	1	2	0	1	0	0	0	1	0	1
	Scissor lift power pack issue	0	0	0	0	0	0	0	0	0	0	0	0
	Scissor lift tilt issue while operating upward and downward	0	1	0	0	0	2	1	0	0	0	0	0
	Lift Downward push button not working	1	1	1	0	1	1	0	0	0	1	0	1
	Lift not working in downward direction	2	2	1	2	3	2	3	1	2	3	1	1
	Dynol application pump issue	0	0	0	0	0	0	0	0	0	0	0	0

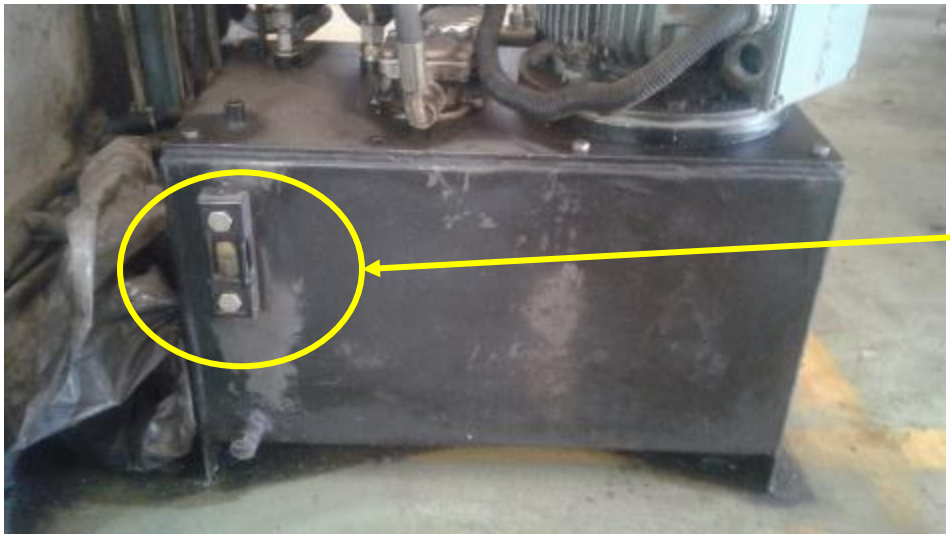
Gun Leakage issue

Past Data Checked we observed that gun failure is frequently reported

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

Noz	Probable Cause	Testing method	Testing and Observations	Conclusion
2	Scissor lift tilt issue while operating upward and downward	Operate lift upward and downward and check	Observed found oil level is less in power pack so top up oil in power pack at required level	Hypothesis invalid



Power pack oil level

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
4	Lift not working in downward direction	Operate lift upward and downward and check	Checked by operating lift but observed 40% of stoppage of lift due to lift not working in downward direction	**Hypothesis valid**

Equipment Name	Possible Causes	19-Apr	19-May	19-June	19-July	19-Aug	19-Sep	19-Oct	19-Nov	19-Dec	19-Jan	19-Feb	19-Mar
	New Operator	0	0	0	0	0	0	0	0	0	0	0	0
Dynol application	Untrained operator	0	0	0	0	0	0	0	0	0	0	0	0
	SOS not followed	0	0	0	0	0	0	0	0	0	0	0	0
	SOS inadequate	0	0	0	0	0	0	0	0	0	0	0	0
	Dynol chemical mixture tank issue	0	0	0	0	0	0	0	0	0	0	0	0
	Dynol lift power pack motor issue	0	0	0	0	0	0	0	0	0	0	0	0
	Dynol gun issue	2	3	1	2	0	1	0	0	0	1	0	1
	Scissor lift power pack issue	0	0	0	0	0	0	0	0	0	0	0	0
	Scissor lift tilt issue while operating upward and downward	0	1	0	0	0	2	1	0	0	0	0	0
	Lift Downward push button not working	1	1	1	0	1	1	0	0	0	1	0	1
	Lift not working in downward direction	2	2	1	2	3	2	3	1	2	3	1	1
	Dynol application pump issue	0	0	0	0	0	0	0	0	0	0	0	0

Last 1 Year Data analyze and Categories in different causes

Past Data Checked we observed that lift not working failure is frequently reported

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
5	Lift Downward push button not working	Check lift operating pendent by operating lift	Checked for 2-3 cycle found ok also operating buttons provided on panel, so if there is issue in pendent then lift will operate from panel button	**Hypothesis invalid**

Lift Operating push buttons not working

Parallel push buttons provided to operate lift

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
7	Dynol application pump issue	Check operating pressure bar for dynol pump is as per specification or not	Checked working pressure is 1.5-2 bar so pump is work properly.	**Hypothesis invalid**

Pump Working Pressure 1.5-2 bar

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Conclusion
1	Dynol gun issue	Check gun spray and no gun leakage	**Hypothesis valid**
2	Lift not working in downward direction	Operate lift upward and downward and check	**Hypothesis valid**

2 Valid Cause

WHY - WHY Analysis

5.3 Identify the root causes:

No	Valid Probable Cause	WHY	WHY	WHY	ACTION / COUNTERMEASURE
1	Dynol gun not working	Spary not coming from Gun Nozzle	Dynol chemical gets leak from center hole instead of tip	Center hole dia more against requirement	Optimize the size
2	Lift not working in downward direction	Lift safety stopper limit switch not operate	Lift safety stopper not properly open	Safety mechanical stopper setting disturb	New safety stopper installed.

6. Implement the corrective measures

Possible Remedies Identified

Three possible solutions identified to eliminate Dynol application Machine not working issue.

1. Reduce pattern size of dynol gun by replacing tip of gun.
2. New safety stopper installed.
3 Spare gun provided on the stage.

Legends

High ↑ Low ↓

S N	Root Cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
1	**1) Dynol gun not working** **2) Lift not working in downward direction**	Spare gun provided on the stage	High ↑	High ↑	No	High ↑	**Rejected**
		Reduce pattern size of dynol gun by replacing tip of gun	Low ↓	High ↑	No	Low ↓	**Accepted**
		New safety stopper installed.	High ↑	Low ↓	Yes	Low ↓	**Accepted**

6. Implement the corrective measures

Possible Remedies Identified

2. Reduce pattern size of dynol gun by replacing tip of gun

CURRENT SITUATION

As dynol application gun tip and its pattern size is more
Before = 525 Size

PROPOSED SITUATION

Dynol application gun tip is replaced with reduced pattern size.
After = 325 Size

CONCLUSION :- This Solution will avoid gun leakage and dynol chemical not spread on vehicle and the work of associate to clean vehicle is reduced .

PROPOSAL ACCEPTED

6. Implement the corrective measures

Possible Remedies Identified

2. New safety stopper installed.

CURRENT SITUATION

Lift mechanical safety stopper not working properly its setting get disturb every time as it depend son scissor lift working operation

PROPOSED SITUATION

New safety stopper is installed which not depend upon scissor lift working operation hence not required to do setting of mechanical safety stopper

CONCLUSION :- This Solution will avoid the safety concern regarding scissor lift tilt and also reduced MTTR.

PROPOSAL ACCEPTED

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
1	Dynol application gun issue	Reduce pattern size of dynol gun by replacing tip of gun .	18.06.2019	Sagar Kekade	CLOSED

PROPOSED SITUATION

Spray tip pattern size of existing gun replaced with small pattern size tip so gun leakage and issue of dynol spread on vehicle is resolved.

6. Implement the corrective measures

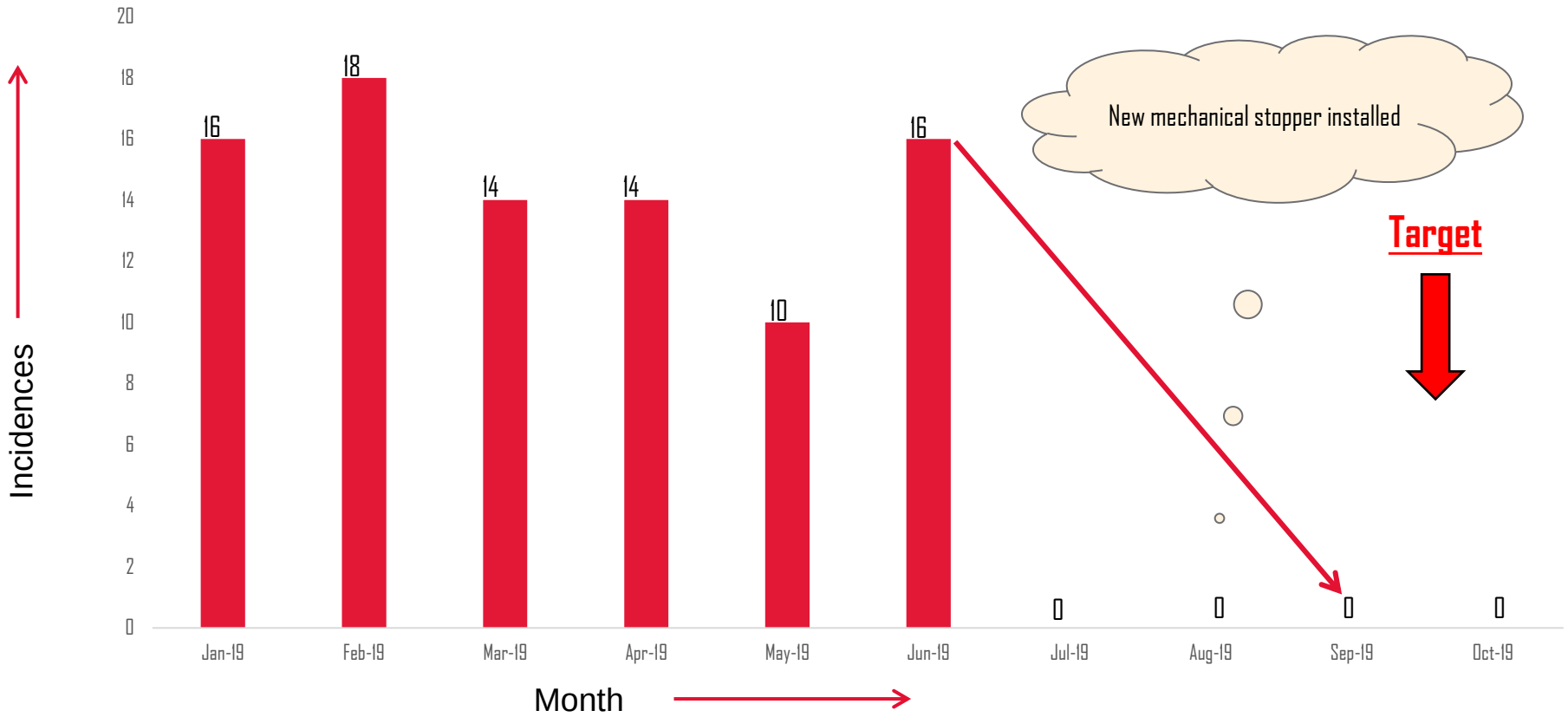
6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
2	Lift not working in downward direction	New safety stopper installed.	18.08.2019	Sagar Kekade	CLOSED

New safety stopper installed
function:- when vehicle lifted upward for dynol application then safety pin require to put in slot on both side of safety stopper. After completion of dynol application, operator downward the lift but if safety pin is not removed by operator then lift will not operate in downward and it will give continuously alarm to operator to remove safety pin on both side of stopper.

7. Check & confirm the results & effects

7.1 Verify Results:



INFERENCE :- **As per data collected ,after action implementation no defect reported in last 5 months. Hence concern is eliminated after taking PCA action through MYB project.**

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	Actual date	Responsibility	Status
1	Design of new reduce size of pattern tip	18.06.2019	Sagar Kekade	CLOSED
2	Installation of new safety stopper	24.09.2019	Sagar Kekade	CLOSED
4	MP Sheet	09.10.2019	Hari Nagar	CLOSED
5	PM Sheets	8.10.2019	Sagar Kekade	CLOSED
6	Daily Check sheet	28.10.2019	Sagar Kekade	CLOSED

Check & confirm the results & effects (Cont.....)

7.3 Tangible & Intangible benefits:

Tangible Benefits -

1) This reduces the cost of new gun which is required to give for spare.
2) Eliminated safety concern regarding scissor lift tilt also it provide safety for the associate .

In tangible Benefits

1) Operator efforts to clean vehicle after dynol application is reduced
2) No vehicle loss

8. Standardization

8.1 Updation of documents: PM Sheets

SN	Document	Actual date	Responsibility	Status
1	PM Sheets	12.08.2019	Sagar Kekade	CLOSED

8.2 Training:

SN	Training Identified	Actual date	Responsibility	Status
1.	Training given to Maintenance Associate.	12.09.2019	Sagar Kekade	Closed

8.3 Additional Audit & check

1	Check point added in daily CAI Checksheet	5.09.2019	Sagar Kekade	Closed

8. Standardization

8.1 Updation of documents: e.g. MP Sheet, PM Sheets, History card.

SN	Document	Actual date	Responsibility	Status
2	PM Check Sheet Updated	12.08.2019	Sagar Kekade	CLOSED

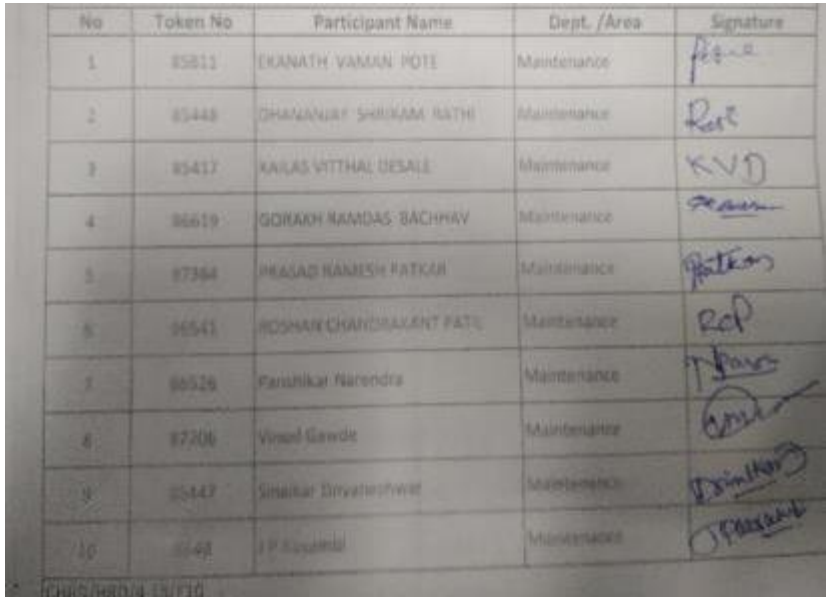
Area :- Test Line | Name of Machine :- Dynol Booth & Lift | Date :

r.No	Item (What)	Activity	Method (How)	Required Value	C/ Nc	Remark
1	Start switch,Main incoming supply	Check/ Tighten	Physical	check the working of s/w,All connection should be tighten	○	
2	Lift Mechanical stopper condition (Rack and pin on type)	Inspect	Visual	Should be working/ No abnormal noise Check stopper bush,all pins	○	
3	Lift Mechanical stopper condition (Roller and guide rollers)	Inspect	Visual	Should be working/ No abnormal noise All pins, guide roller	○	
4	Lift Mechanical stopper condition (Electrical)	Check	Visual	All limit switch should be working, faster working	○	
5	Lamps & push buttons / Selector switches	Inspect	Visual	Ensure All emergency switches are working Ok, Selector switch & indicators are working Ok.	○	
6	Abnormal sound of any assembly / component (electrical Panel / air blower / Heater etc.)	Inspect	Listen	No abnormal sound	○	
7	Application Gun	Clean	Visual / task	Ensure that the gun is clean , no leakage of the fluid & the pattern setting is Ok	○	
8	All area	Clean	Physical with cloth & brush	Clean Inside area & components of the panel including all poles,hydraulic motor,Trolly ,	○	
9	Pump/ Motor mounting	Tighten	Physical	Tight the mounting bolts of motor	○	
10	Pump/Motor temperature	check	physical	should be below 70° C	○	
11	Oil	Check	Physical	Check oil condition if not ok replace & maintain oil level	○	
12	Lift cylinder	Check/ Tighten	Physical	Lift and cylinder locking pin,all bush,Lift rollers	○	
13	Hose joints	Check/ Tighten	Physical	Cylinder joints tighten for no leakage	○	

8. Standardization

8.2 Training:

Action Plan	Target Dt	Resp.	Status
Training given to Maintenance 11 Associate.	12.09.2018	Sagar Kekade	Closed



No	Token No	Participant Name	Dept. /Area	Signature
1	85811	EKANATH VAMAN POTE	Maintenance	
2	85448	DHANANJAY SHRIRAM RATHI	Maintenance	
3	85417	KAILAS VITTHAL DESALE	Maintenance	KVD
4	86619	GORAKH RAMDAS BACHHAV	Maintenance	
5	87384	PRASAD RAMESH PATKAR	Maintenance	
6	86541	ROSHAN CHANDRAKANT PATIL	Maintenance	
7	86526	Panshikar Narendra	Maintenance	
8	87206	Vinod Gawde	Maintenance	
9	85447	Sinaikar Divakershwar	Maintenance	
10	85[illegible]	[illegible]	Maintenance	

8. Standardization

8.3 Additional Audit & check

CAI daily check sheet updated

Shower QDMS And Dignostic	1	Scanner & PC	VISUALLY	Scanner & PC should working properly	
	2	Dignostic Testor line and shower	VISUALLY	OBD cable and its PC should working OK	
	3	Dignostic Warap side	VISUALLY	OBD cable and its PC should working OK	
	4	Dignostic Major mechanical	VISUALLY	OBD cable and its PC should working OK	
ECOS	1	Client 1	CHECK	1.Statrt PC & Printer 2. Check Probe communication i.e. green light on probe and axis point should blink continiously after PC Start 3.Proble cable and clamp Red and Black is in good condition. 4. Check 1 Cycle that ECOS working OK 5.Check Wi-Fi router continuously ON	
	2	Client 2	CHECK	1.Statrt PC & Printer 2. Check Probe communication i.e. green light on probe and axis point should blink continiously after PC Start 3.Proble cable and clamp Red and Black is in good condition. 4. Check 1 Cycle that ECOS working OK	
	3	Repair	CHECK	1.Statrt PC & Printer 2. Check Probe communication i.e. green light on probe and axis point should blink continiously after PC Start 3.Proble cable and clamp Red and Black is in good condition. 4. Check 1 Cycle that ECOS working OK	
	4	*Servar And Calibration unit*	CHECK	1.Statrt PC 4. Check Calibration Unit it is in working condition.	
Wheel Nut	1	*Torque tool*	CHECK	Both Torque tool in working condition	
	2	*Scanner*	CHECK	Scanner working properly	
	3	*machine*	CHECK	Start machine and observe 1 cycle for OK printing	
Smoke	1	*Machine*	CHECK	Start machine and observe 1 cycle for OK printing	
	2	*Machine chember*	Clean	Clean both glass of machine, also clean chember.	
Dent puller	1	*Machine*	CHECK	Both machine working OK no loose connection	
CO2 Machine	1	*Machine*	CHECK	Check machine condition for loose connetion	
Acetylene	2	*Acetylene Cylinder*	CHECK	Check cylender for leakage and gauge condition.	
Paint Booth	1	*Heater and heater Gun*	CHECK	All heters and heter Gun in working condition	
Engine Hoist	1	*Hoist*	CHECK	Hoist in working condition and no damage push button	
R Pass & Headlamp	1	R Pass & Headlamp Aiming	CHECK	Both R- Pass & Head Lamp Aiming is in smooth working condition	
CAI Lift	1	*Lift*	CHECK	Lift working ok	
	1	*Final Lift*	CHECK	1.Lift working Ok. 2. Oil level OK. 3.No oil leakage	

Page 2

9. Summarize & Plan future actions

9.1 Conclusion:

With implementation of new safety stopper the safety concern regarding scissor lift is eliminated.

9.2 Horizontal Deployment:

Input given For 4 post lift.

9.3 Lessons Learnt:

Things Gone Wrong : Initially at the time of installation of new safety stopper time to insert pin in stopper is not considered.

Things Gone Right :

Detailed analysis on past data help us to resolved the completely.

Thank you

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